

ECE 253 – Digital Image Processing

DeHazing/DeFogging

Lecture Overview

- Image Degradation Model
- Image Restoration
- Image Dehazing/Defogging
- Learning-Based Dehazing/Defogging

Image Degradation

- Image degradation due to
 - Noise in transmission
 - Imperfect image acquisition
 - Environmental condition
 - Quality of sensor



Google Image



Original image



Noisy image



Denoised image

Wikipedia

Image Degradation

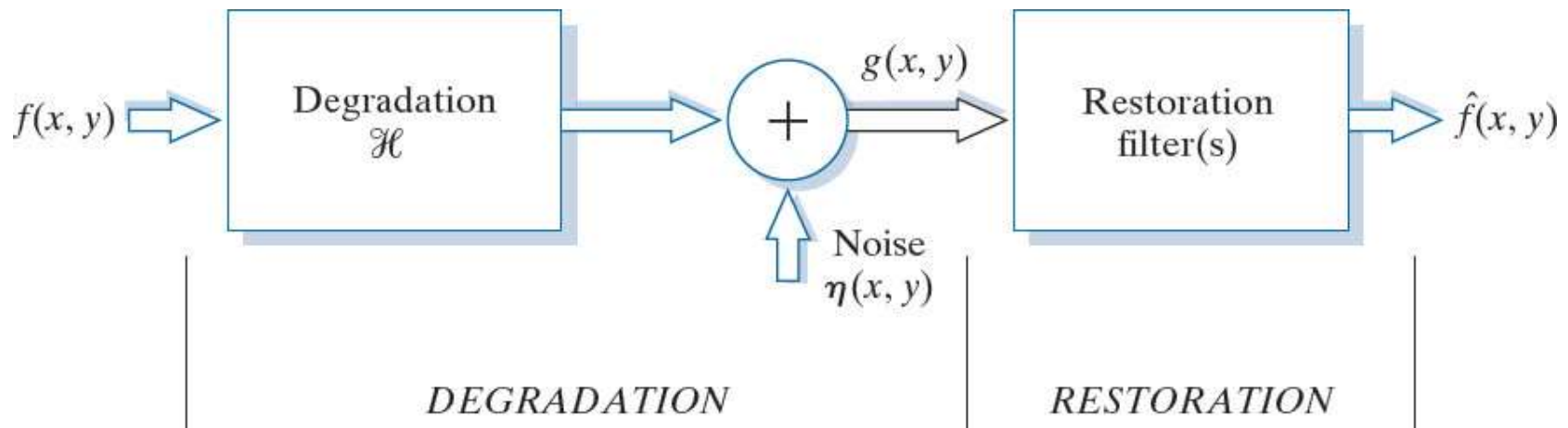
- Can we improve the image, or “undo” the effects?
- **Image Enhancement**
 - “improve” an image subjectively
 - e.g., contrast stretching
- **Image Restoration**
 - Remove distortion from image in order to go back to the “original” => **objective** process.
 - e.g., deblurring, denoising

Image Restoration

- Task of recovering an image from its degraded version assuming some **knowledge of the degradation phenomenon**.
- **Models** the degradation process and **inverts** it to obtain the original from the degraded (observed) image.

Degradation Model

$$g(x, y) = H[f(x, y)] + \eta(x, y)$$



Degradation Model

- $f(x, y)$ – image before degradation, 'true image'
- $g(x, y)$ – image after degradation, 'observed image'
- H – some knowledge about the degradation function
- $\hat{f}(x, y)$ – estimate of $f(x, y)$ computed from $g(x, y)$
- $\eta(x, y)$ – additive noise

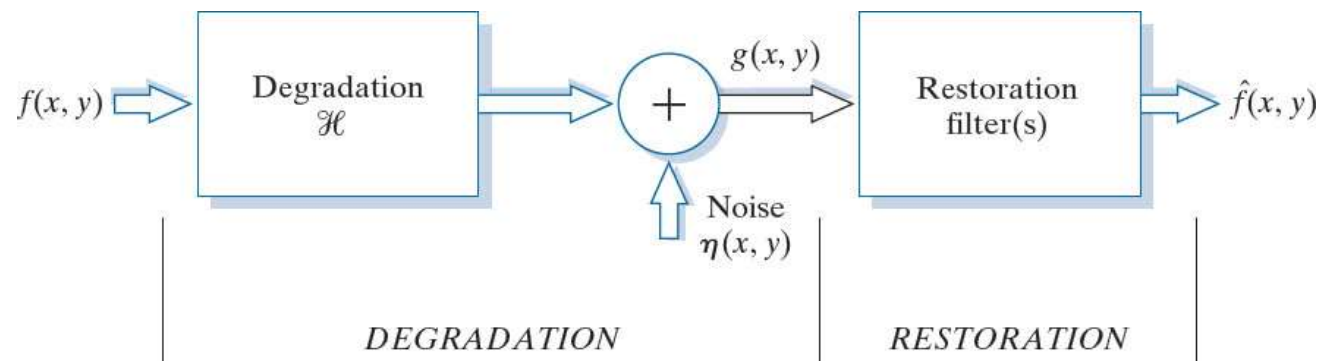


Image Restoration

- Use a priori knowledge of the degradation
- Modeling the degradation and ply the inverse process
- Formulate and evaluate objective criteria of goodness

$$g(x, y) = H[f(x, y)] + \eta(x, y)$$

→ Design restoration mechanism such that $\hat{f}(x, y)$ is as close to $f(x, y)$ as possible

Image Dehazing/Defogging

- A process of restoring images by **removing haze, fog, or mist** to **improve image quality and visibility**
- Dehazing algorithms are designed to
 - Estimate the amount of haze in different parts of an image
 - Apply corrective measures to restore the original scene as closely as possible

Image Modeling in Haze

- Use a priori knowledge of the degradation (**H**)
- Haze is typically caused by atmospheric particles like dust, smoke, or water droplets, which scatter light and reduce the contrast and color fidelity of an image.
- This scattering effect results in images that appear faded, with a loss of detail, especially in distant objects.

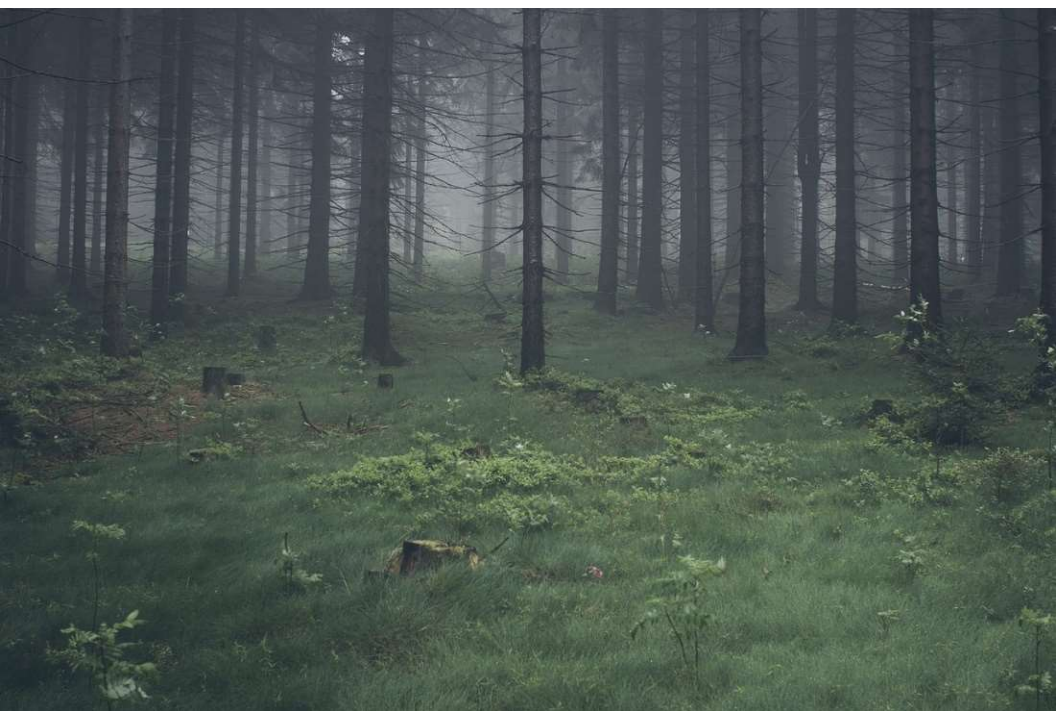
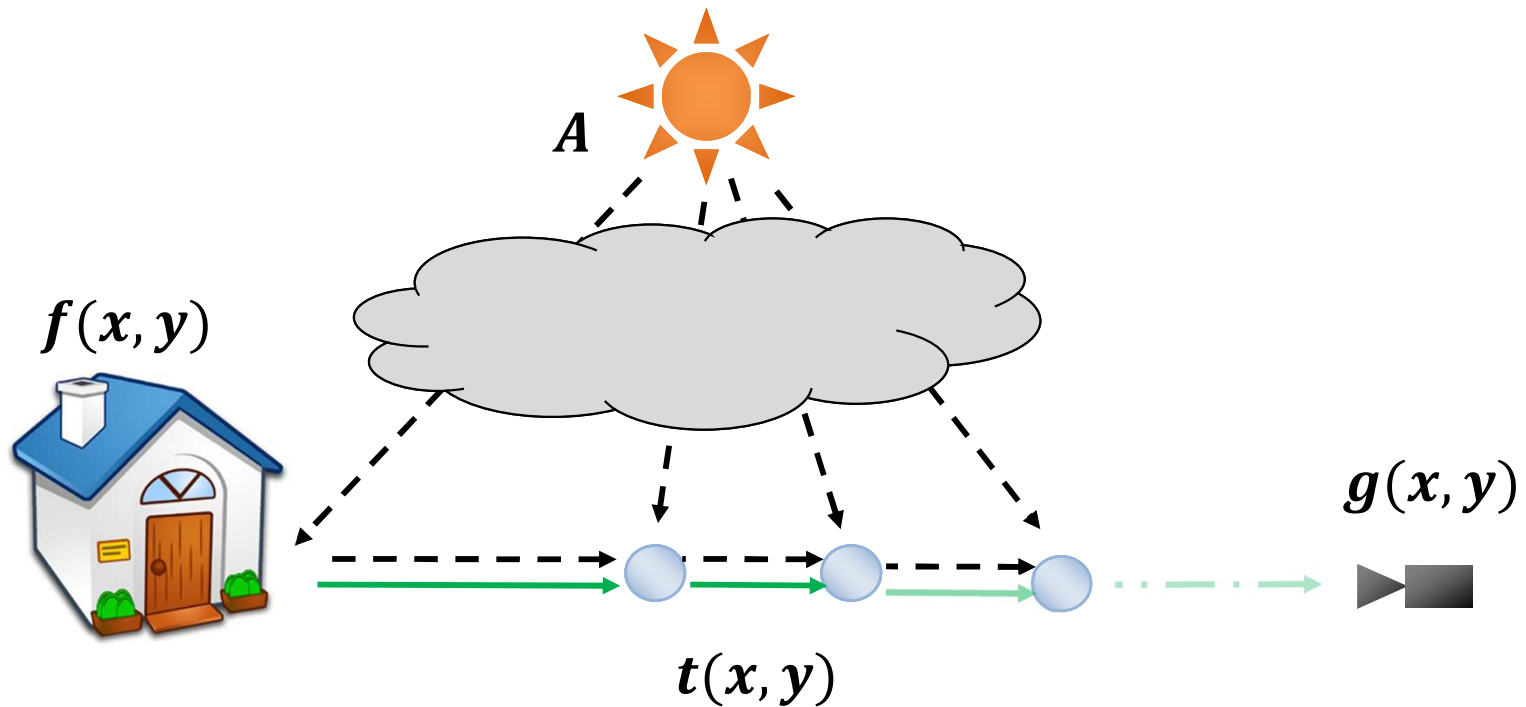


Image Modeling in Haze



$$g(x, y) = t(x, y)f(x, y) + (1 - t(x, y)) \cdot A$$

$f(x, y)$: Hazy-free image

$g(x, y)$: Hazy image

$t(x, y)$: Transmission map

A : Atmospheric light

Image Modeling in Haze

- **Transmission** map is the amount of light that reaches the camera from the scene after passing through the haze.
- It describes the level of haze present at each pixel ranging from [0, 1] in the image, indicating how much of the scene's original color and brightness has been affected by the haze

$$t(x, y) = e^{-\beta \cdot d(x, y)}$$

β : Scattering coefficient

$d(x, y)$: Distance from camera to scene

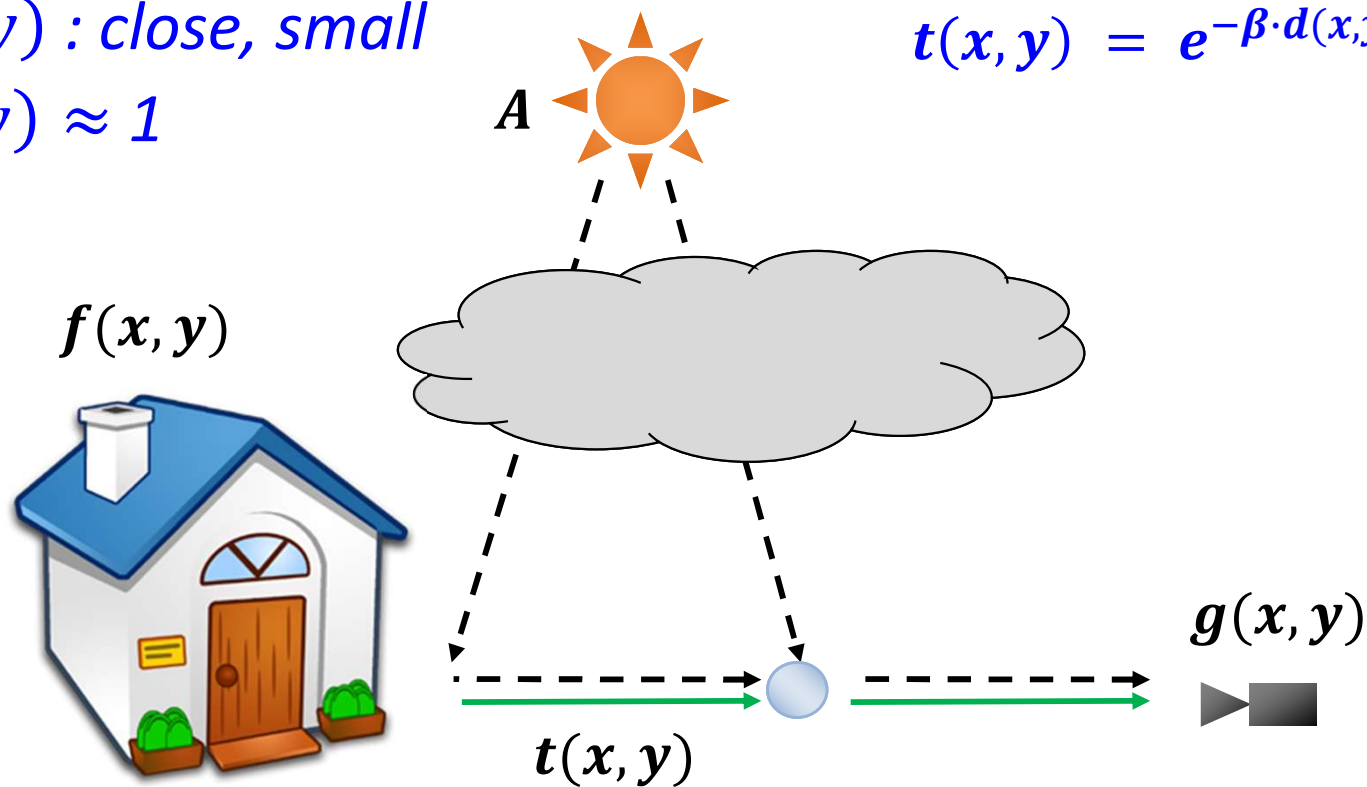
Haze Image Model: Very Close

- Little to no haze, meaning the scene is relatively clear

$d(x, y)$: close, small

$t(x, y) \approx 1$

$$t(x, y) = e^{-\beta \cdot d(x, y)}$$



$$g(x, y) = t(x, y)f(x, y) + (1 - t(x, y)) \cdot A$$

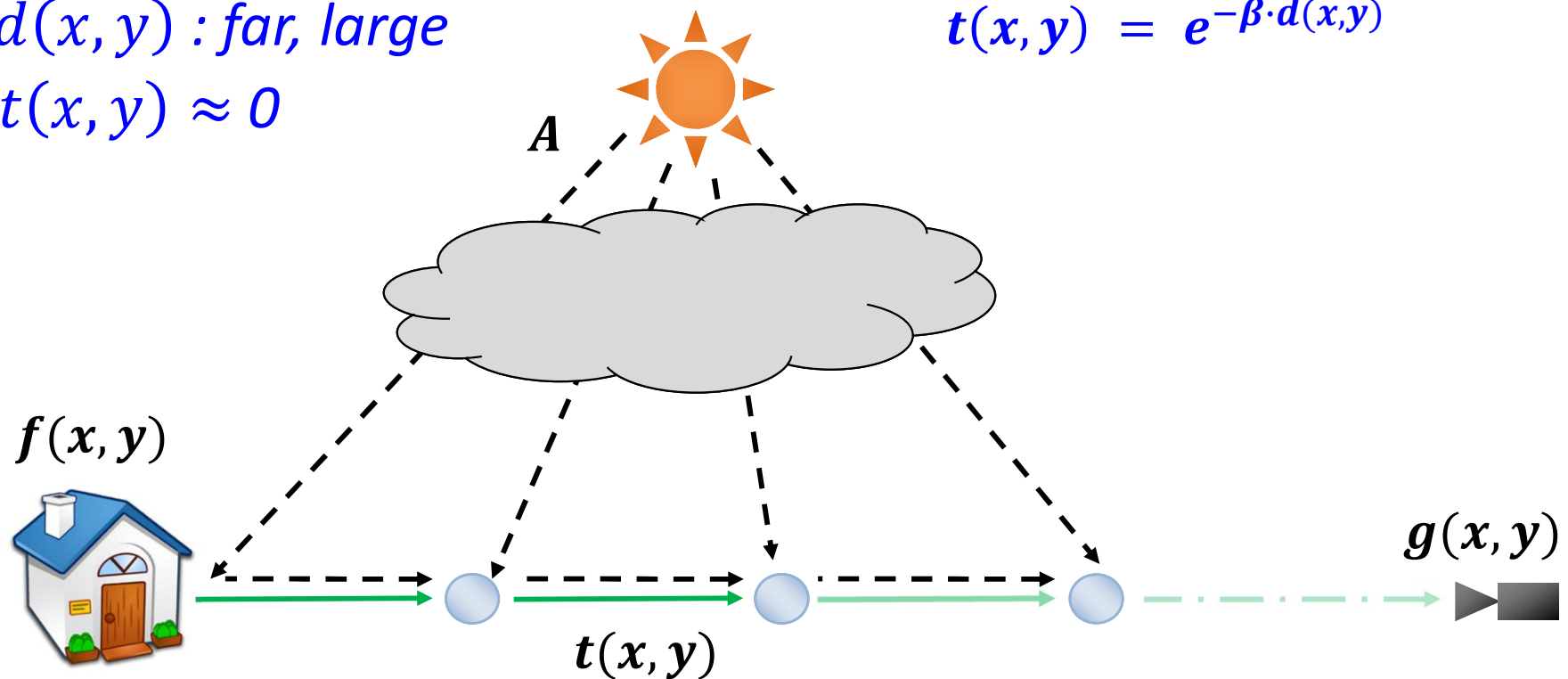
$$g(x, y) \approx f(x, y)$$

Haze Image Model: Far Away

- Heavy haze, meaning the scene is mostly obscured

$d(x, y) : \text{far, large}$
 $t(x, y) \approx 0$

$$t(x, y) = e^{-\beta \cdot d(x, y)}$$



$$g(x, y) = t(x, y)f(x, y) + (1 - t(x, y)) \cdot A$$
$$g(x, y) \approx A$$

Image Dehazing

- Invert the degradation process

$$g(x, y) = t(x, y)f(x, y) + (1 - t(x, y)) \cdot A$$

known



want to
find



Image Dehazing

- **Goal:** To **estimate transmission** and **atmospheric light** to obtain a dehazed image

$$g(x, y) = t(x, y)f(x, y) + (1 - t(x, y)) \cdot A$$

Diagram illustrating the components of the image formation model:

- $g(x, y)$: known (blue arrow)
- $t(x, y)$: estimated (red arrow)
- $f(x, y)$: want to find (blue arrow)
- A : estimated (red arrow)

➡

$$\hat{f}(x, y) = \frac{g(x, y) - \hat{A}}{\max(\hat{t}(x, y), \varepsilon)} + \hat{A}$$

ε : Numerical conditioning (0.001)

Image Dehazing



Hazy: g



Transmission: t



Dehazed: $\hat{f}$



Hazy : g



Transmission: t



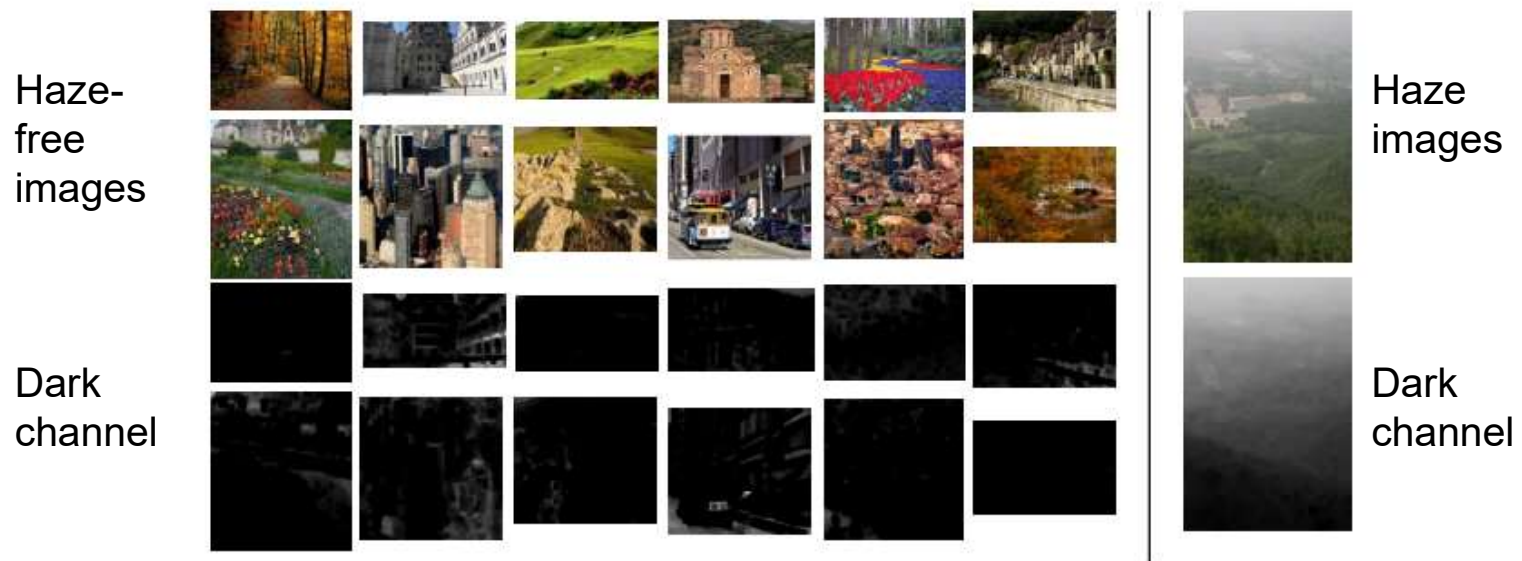
Dehazed: $\hat{f}$

One Approach: Transmission Estimation

- **Dark Channel Prior (DCP)**

- a widely used method based on statistics
- in most natural images, some pixels in the non-sky regions have very low intensity in at least one color channel

- **$\min(\text{rgb}, \text{local patch})$**



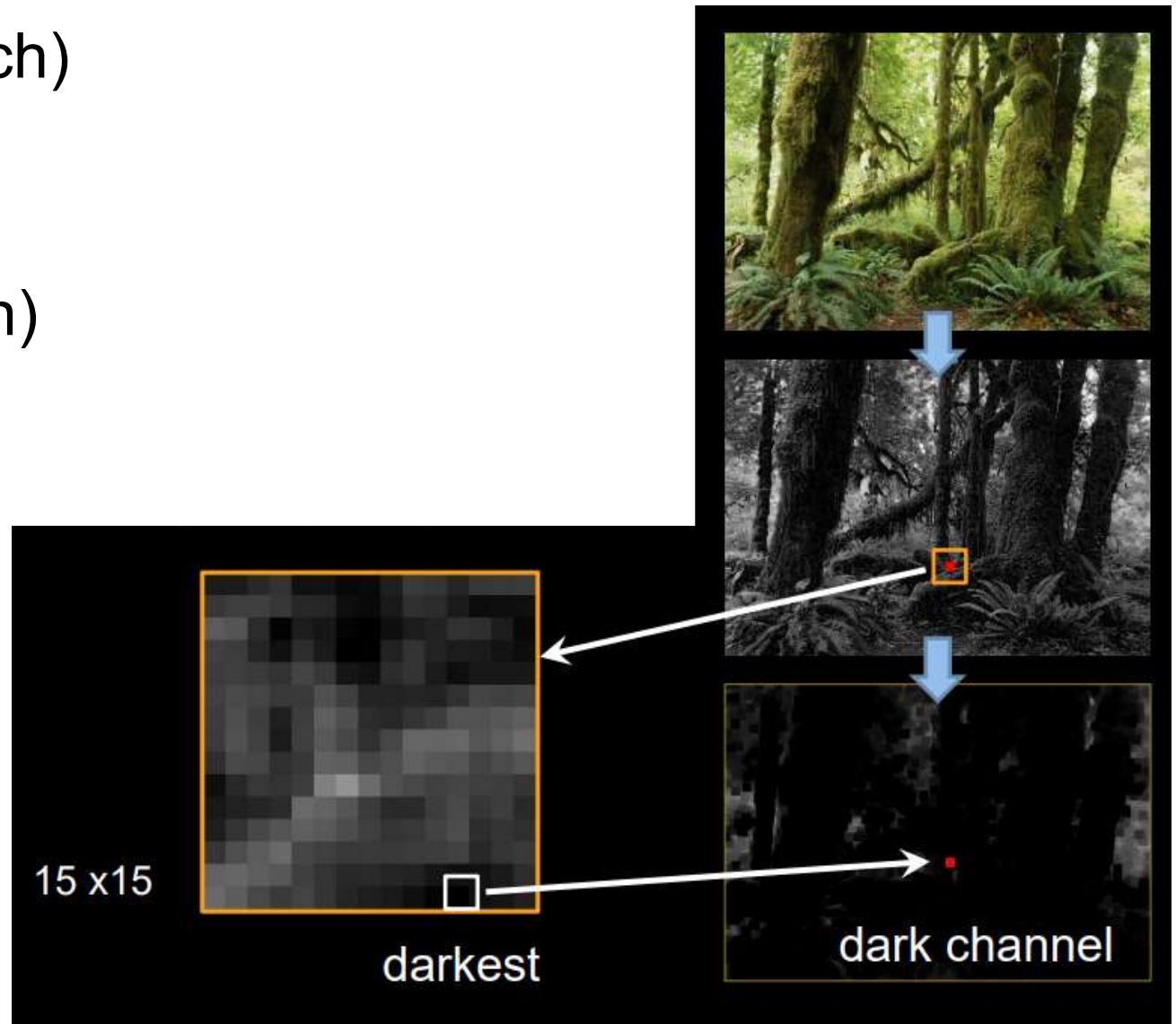
Dark Channel Prior

- $\min(\text{rgb}, \text{local patch})$
- $\min(r, g, b)$



Dark Channel Prior

- $\min(\text{rgb}, \text{local patch})$
 - $\min(r, g, b)$
 - $\min(\text{local patch}) = \text{min filter}$



One Approach: Atmospheric Light Estimation

- Atmospheric light tends to be the **brightest part** of the hazy image
 - pick the top 0.1% brightest pixels in the dark channel and analyze their statistics

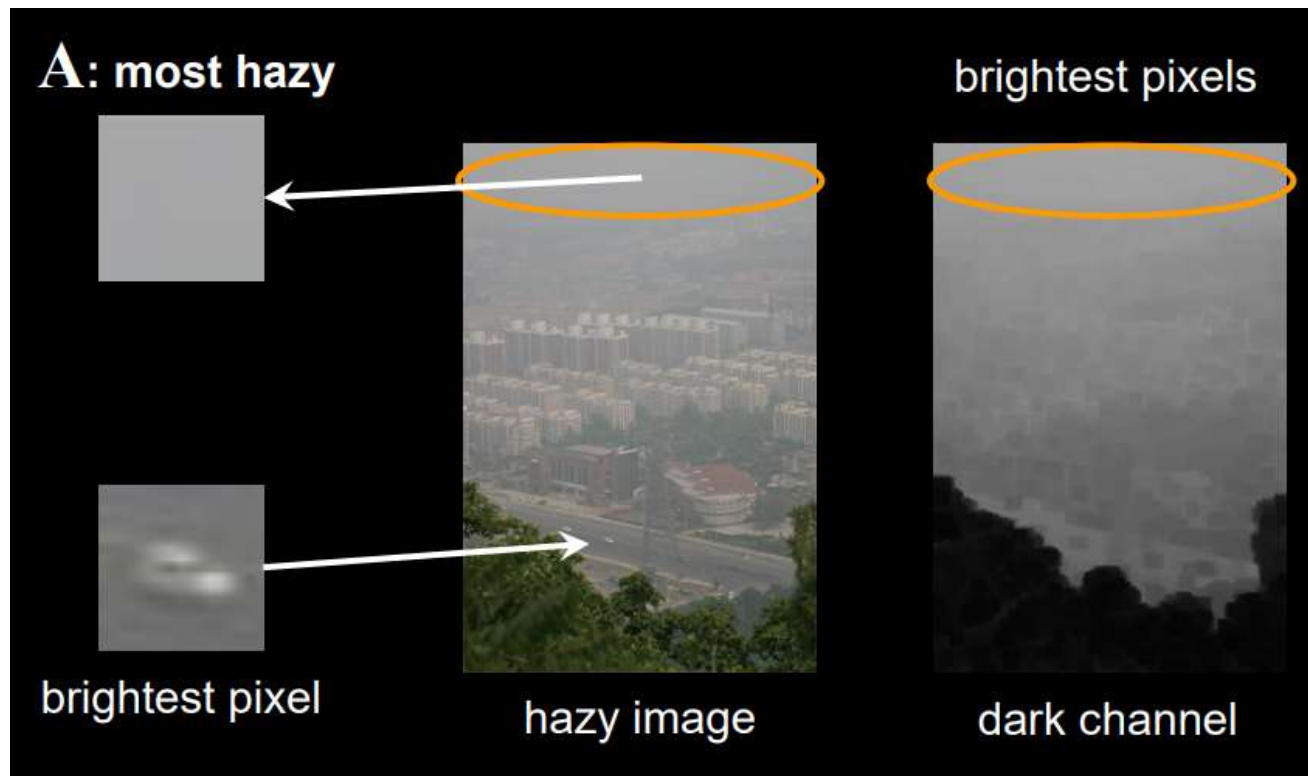
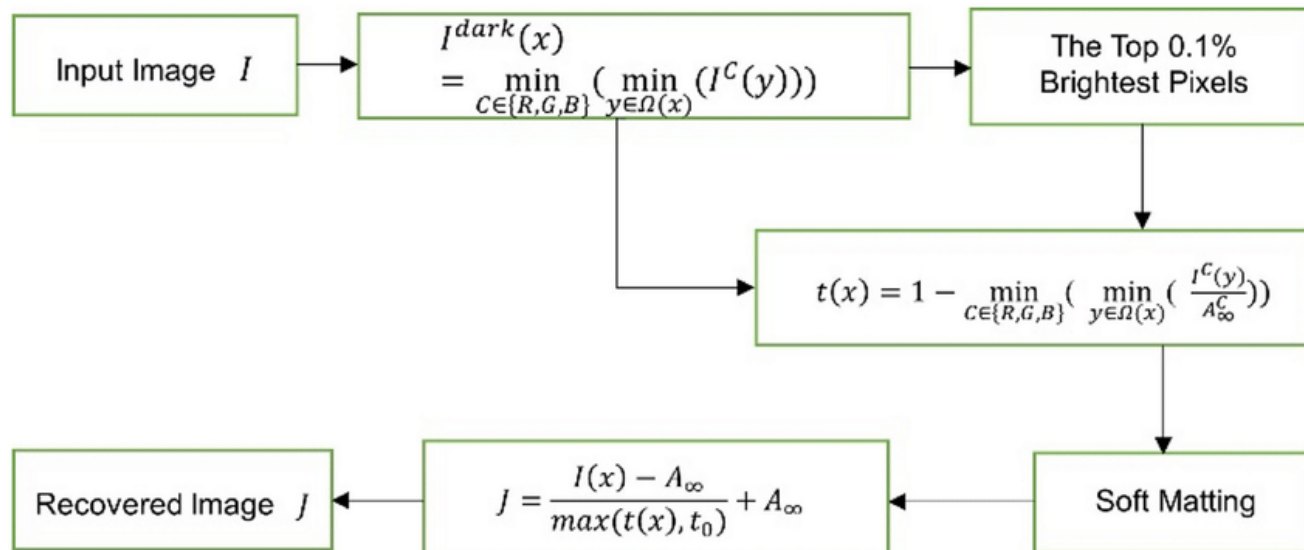


Image Dehazing

- Invert the degradation process using the estimated transmission and atmospheric light

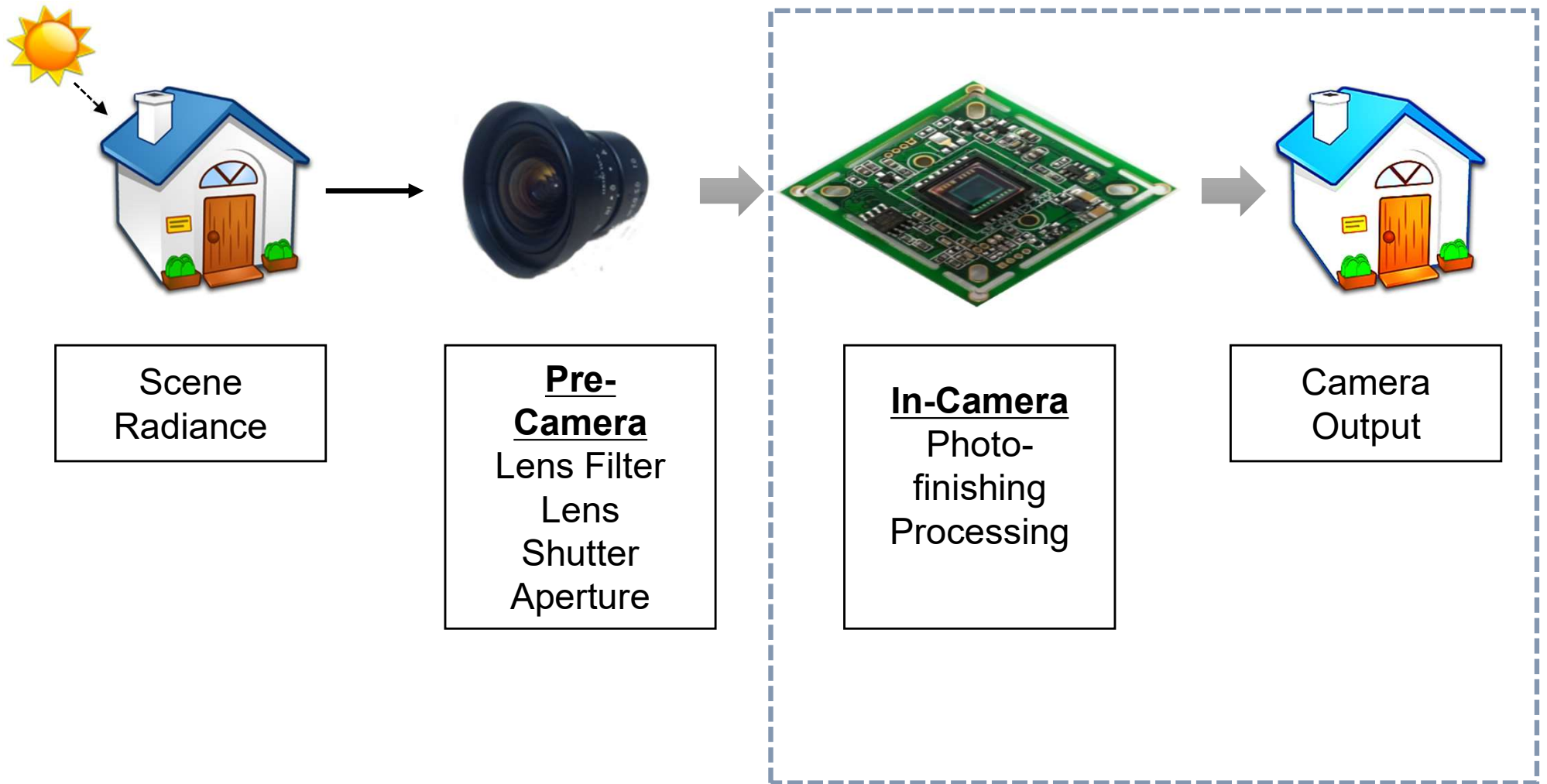
$$\hat{f}(x, y) = \frac{g(x, y) - \hat{A}}{\max(\hat{t}(x, y), \varepsilon)} + \hat{A}$$

ε : Numerical conditioning (0.001)



Joint Defogging and Demosaicking

Photography Pipeline



Camera Processing Pipeline

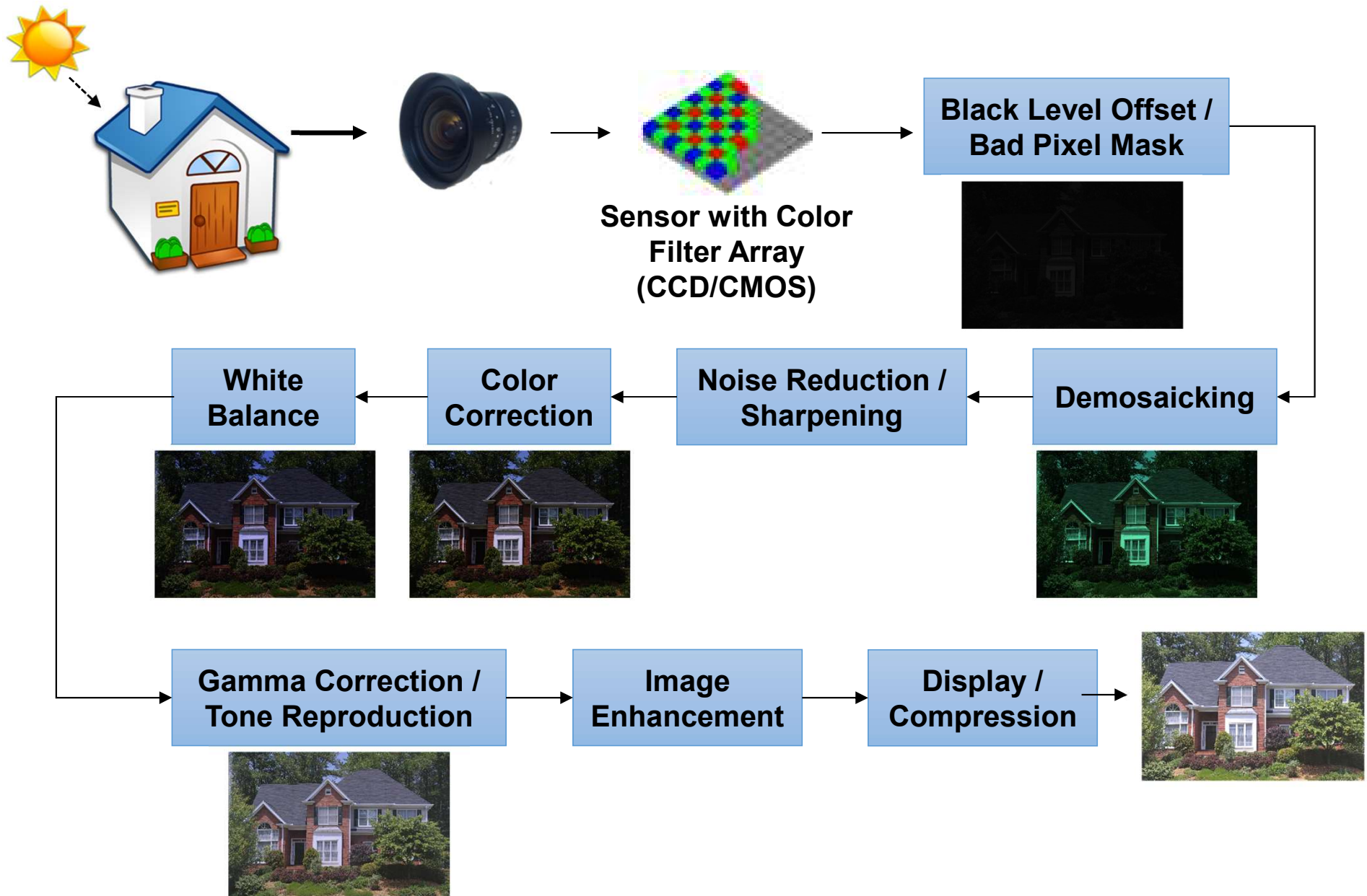
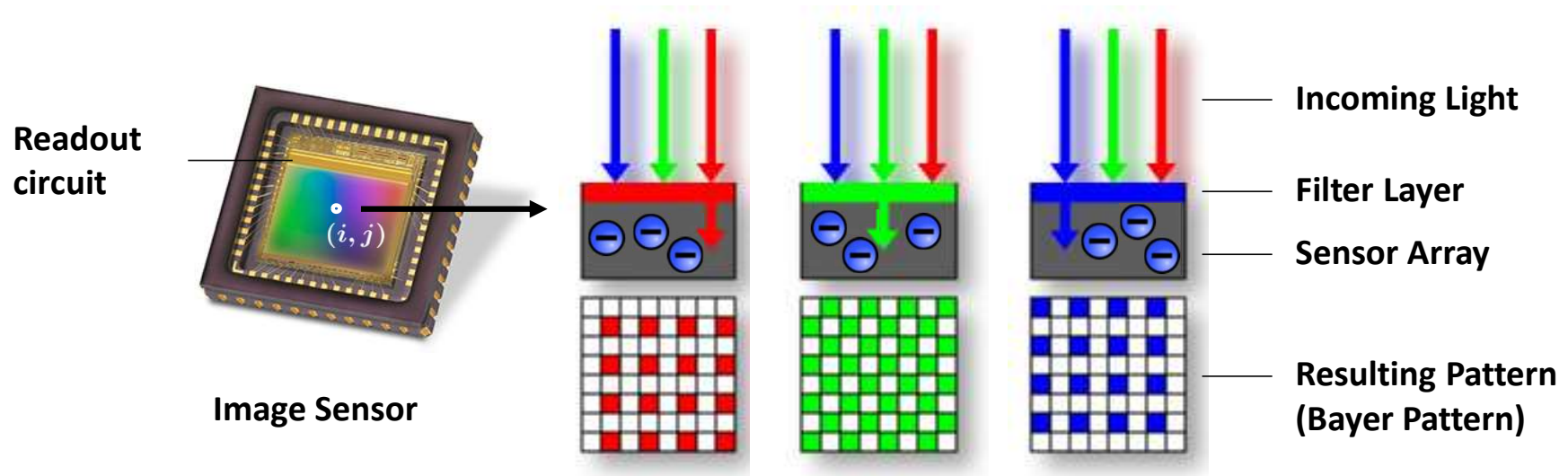


Image Sensor

- Photon-to-electron conversion (random photon arrival)
- Electron-to-voltage conversion/Analog-to-digital conversion (readout circuit)



White Balance

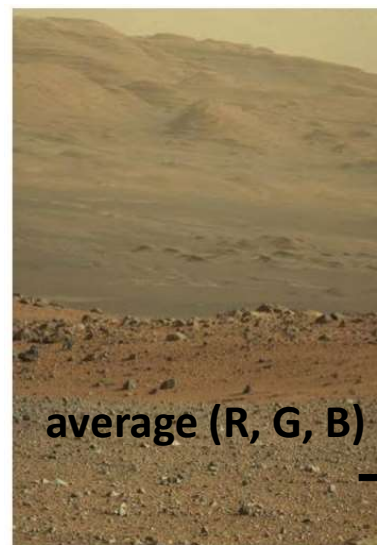
- Global adjustment of the intensities of the colors
- Algorithms
 - Gray world assumption: make average of all pixels gray
 - White world assumption: find brightest region of image and make it white



Unprocessed color



White balanced color
(a gray surface neutral
in the same light)



Unprocessed color



White balanced color
(something in the scene
is white)

average (R, G, B) = (100%, 85%, 59%)
→ (100%, 100%, 100%)

White Balance

Color Temperature	Light Source
1000-2000 K	Candlelight
2500-3500 K	Tungsten Bulb (household variety)
3000-4000 K	Sunrise/Sunset (clear sky)
4000-5000 K	Fluorescent Lamps
5000-5500 K	Electronic Flash
5000-6500 K	Daylight with Clear Sky (sun overhead)
6500-8000 K	Moderately Overcast Sky
9000-10000 K	Shade or Heavily Overcast Sky



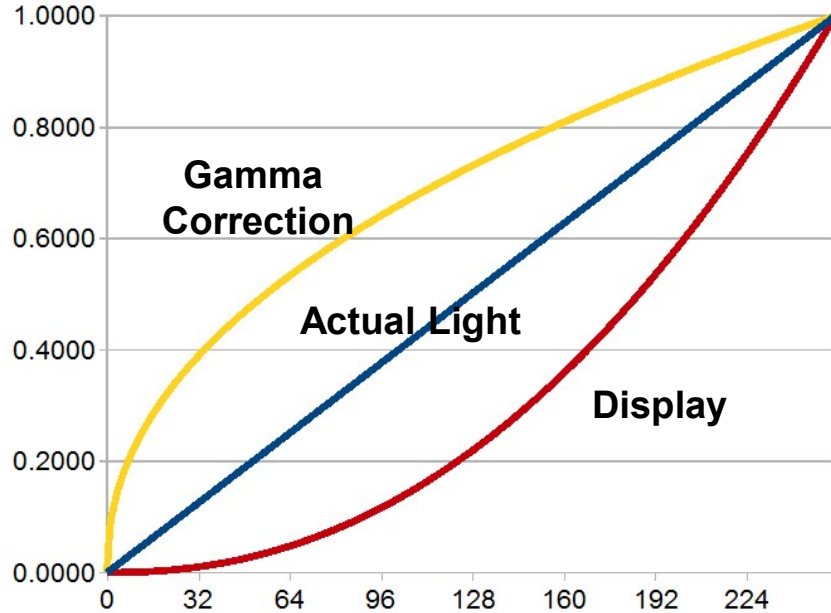
Color Cast



Daylight White Balance

Gamma Correction

- Enhance the contrast of the displayed images



Actual light



With gamma correction



Display



Actual light



Without gamma correction



Display

Demosaicking

- Reconstruct a full color image from sensor images

$$g(i, j) = \phi(i, j)^T \mathbf{x}(i, j)$$

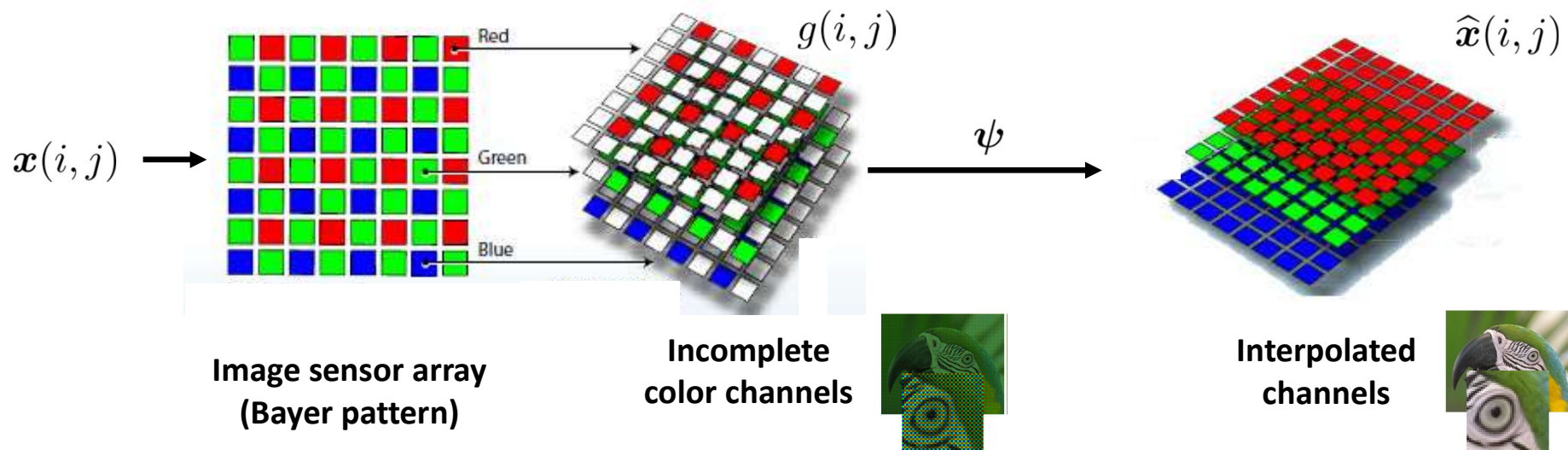
$\phi : \mathbb{Z}^2 \rightarrow \{0, 1\}^3$ (Sampling lattice)

Demosaicking

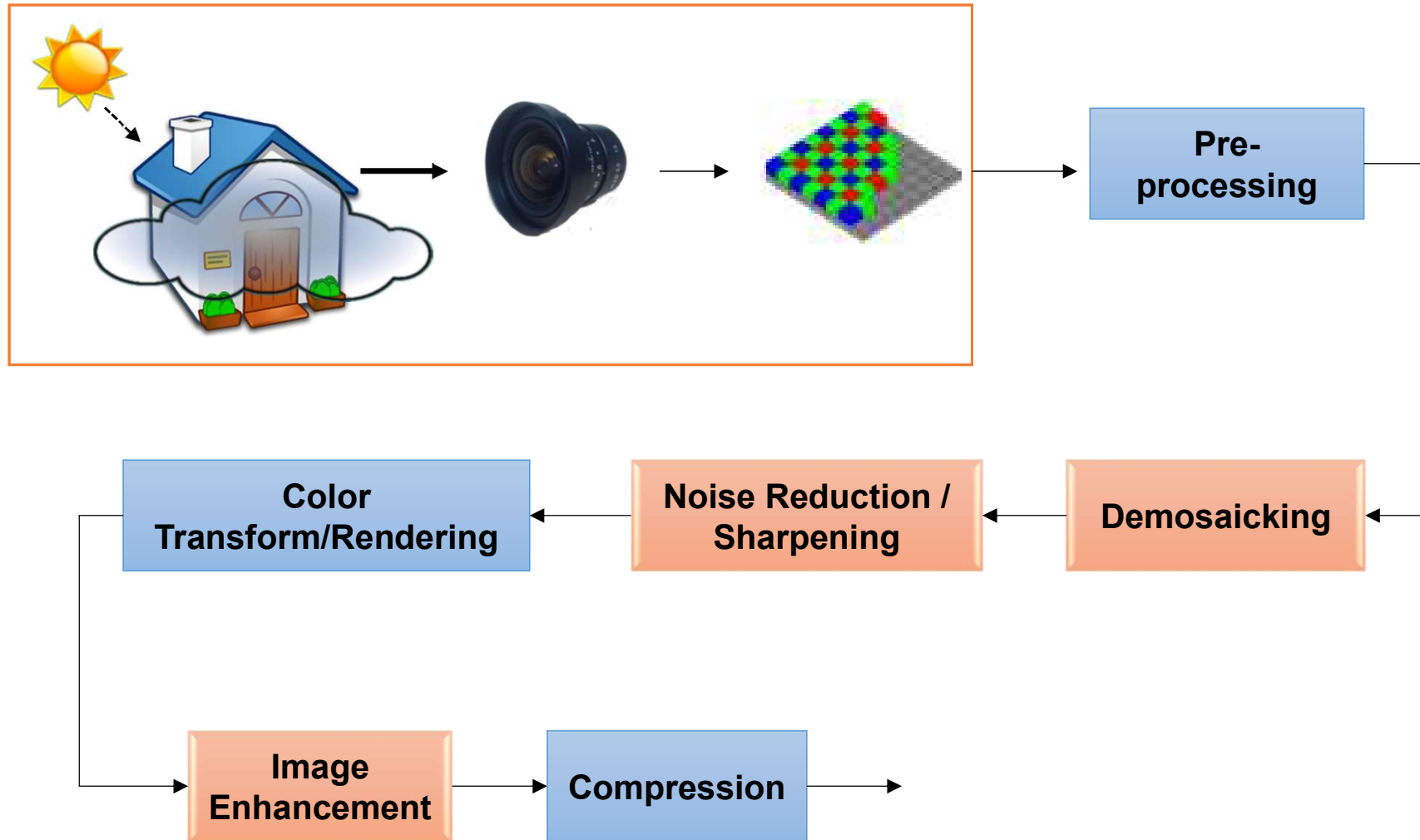
$$\hat{\mathbf{x}}(i, j) = \psi\{g\}(i, j)$$

ψ : Demosaicking operator

e.g. $\phi(i, j) = [1 \ 0 \ 0]^T$ for a red sample, $g(i, j) \in \mathbb{R}$, $\mathbf{x}(i, j) \in \mathbb{R}^3$

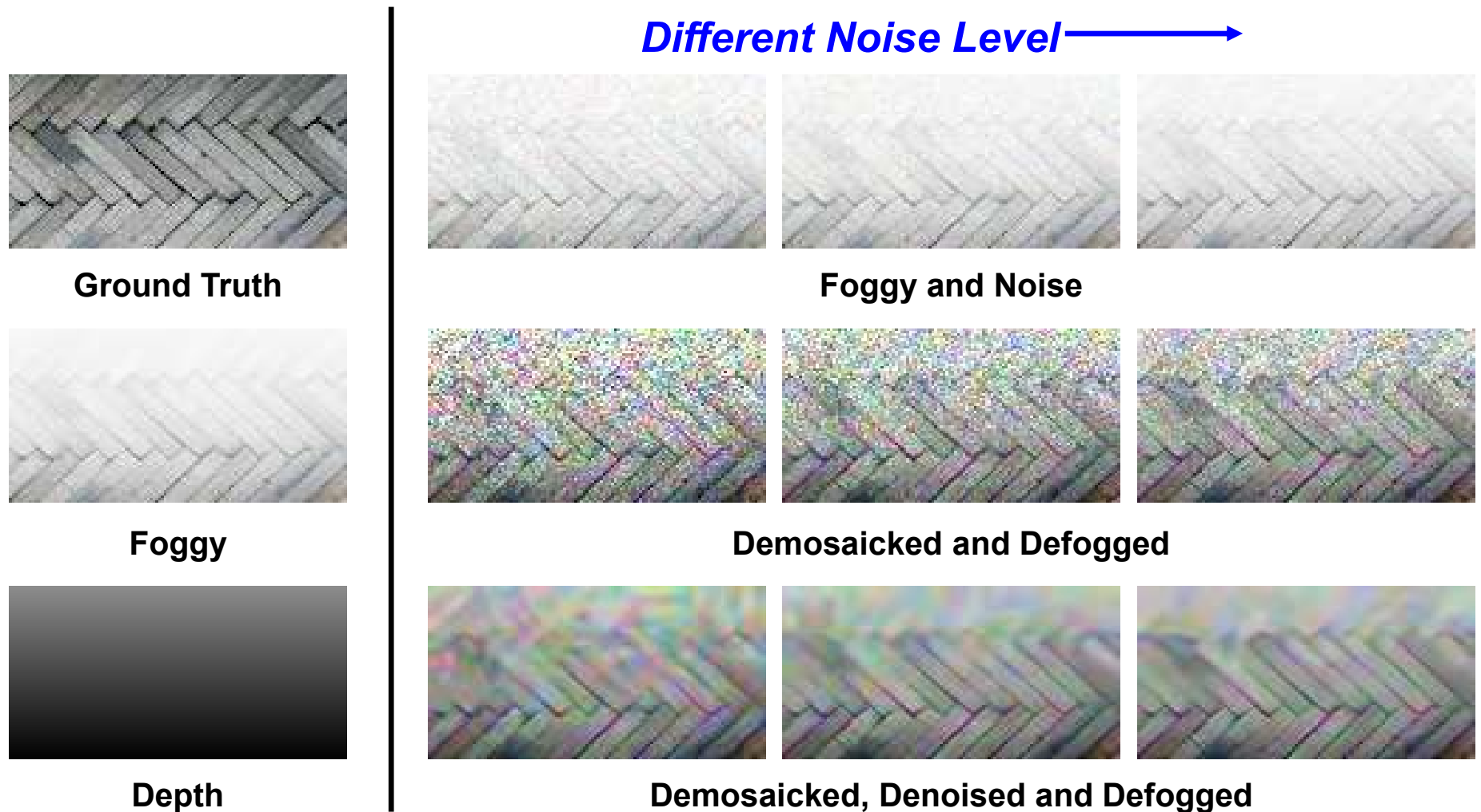


Camera Pipeline – Aware Image Acquisition



Sensor Noise: Simulation

- Noise is amplified by dehazing process.



Demosaicking: Simulation

- Demosaicking artifacts are also amplified by dehazing process.

Without Noise



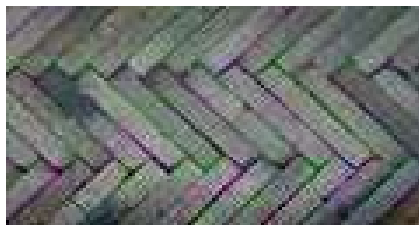
Foggy



Ground Truth



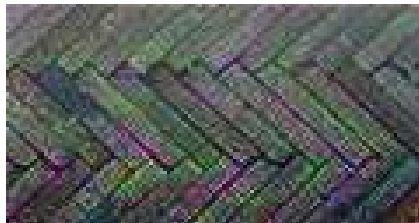
Demosaicked



Defogged



Demosaicked

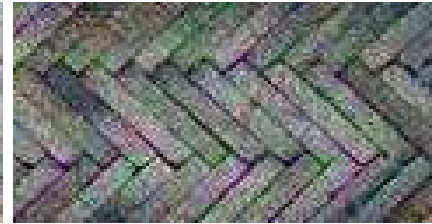


Defogged

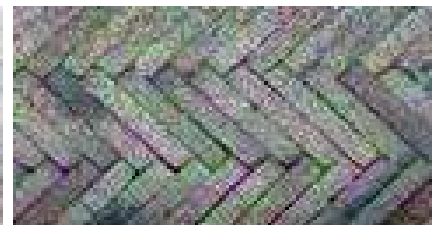
With Noise



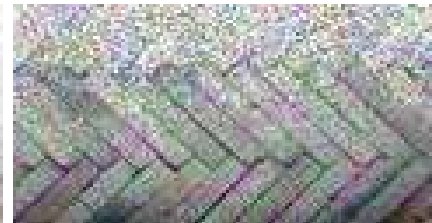
Light Fog



Moderate Fog



Thick Fog

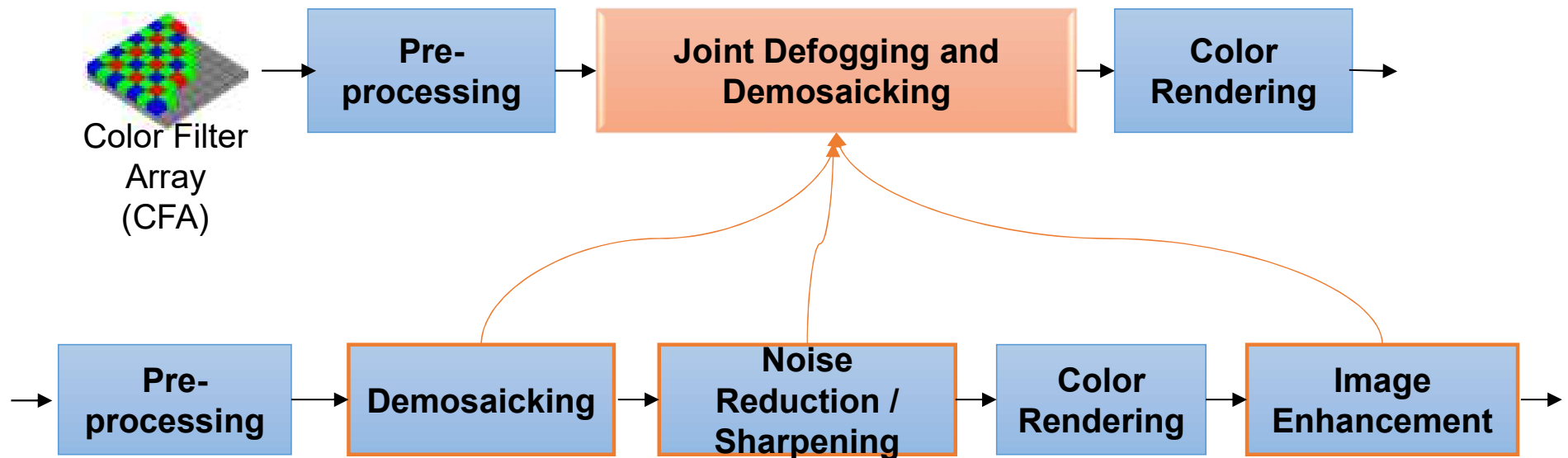


Different Fog Amount



Joint Demosaicking and Dehazing

- We propose an alternative digital camera processing pipeline based on combining demosaicking, denoising, and defogging processes.



Results



Foggy



DCP



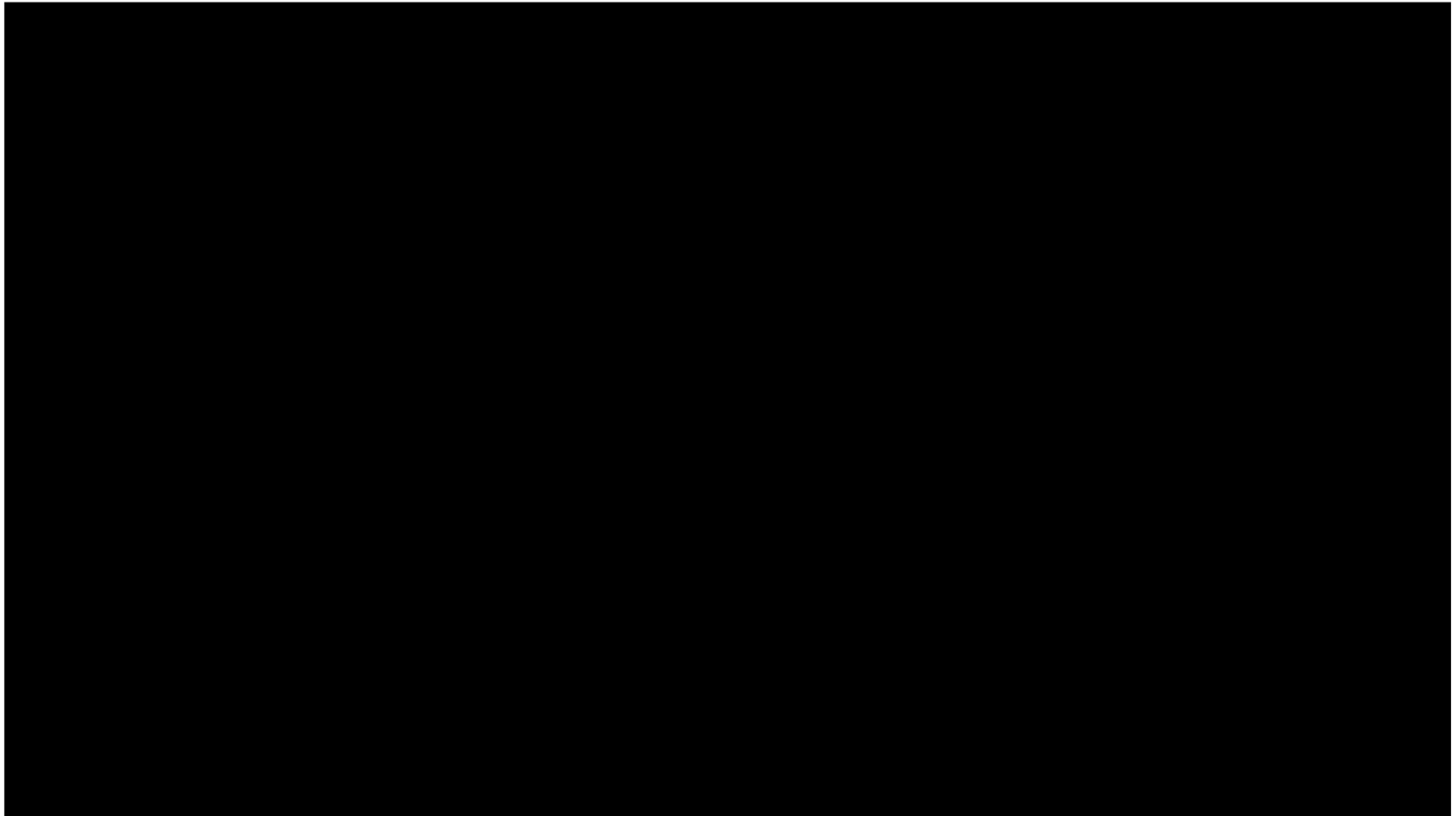
Tarel 's



Proposed

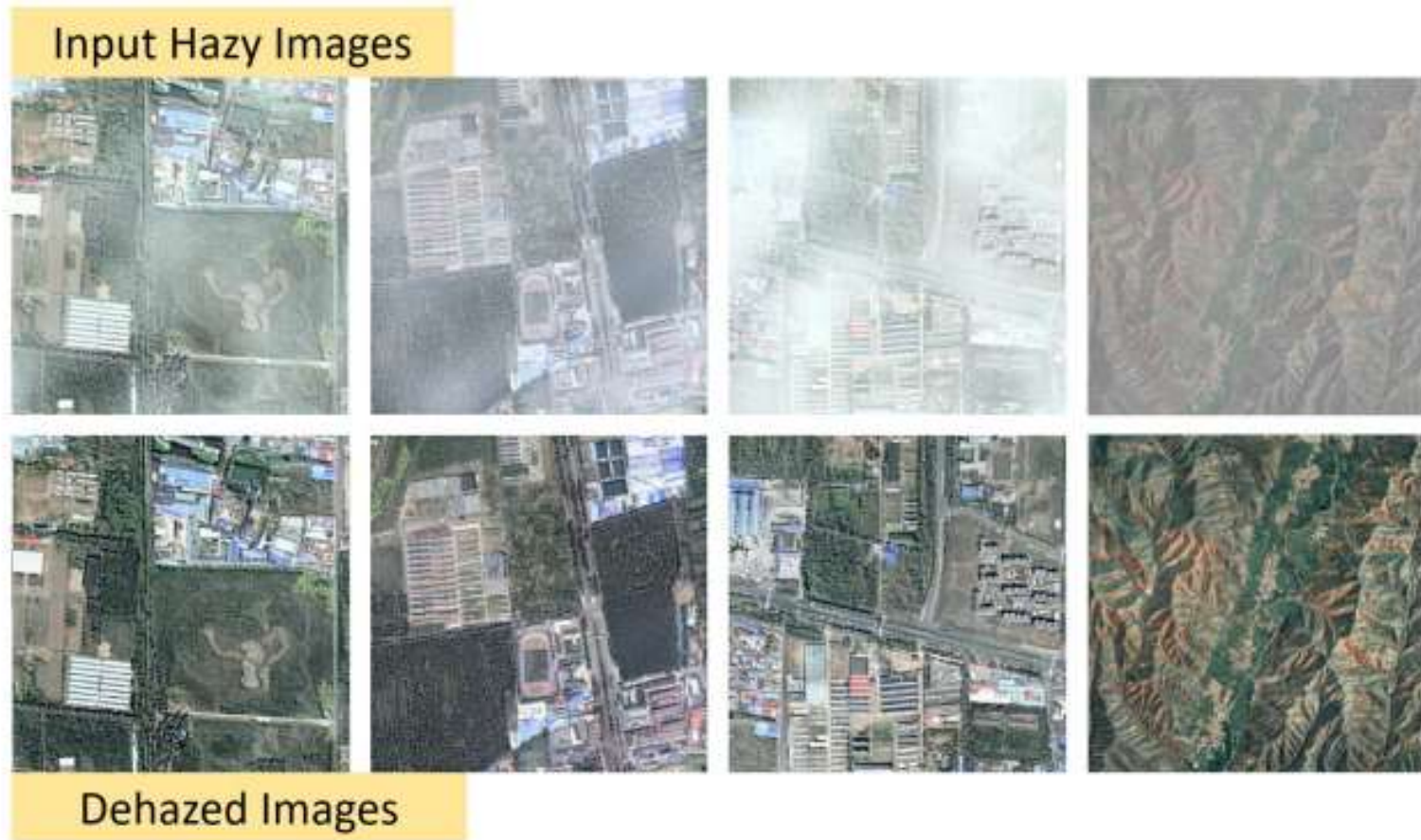
Other Applications

- Video Dehazing



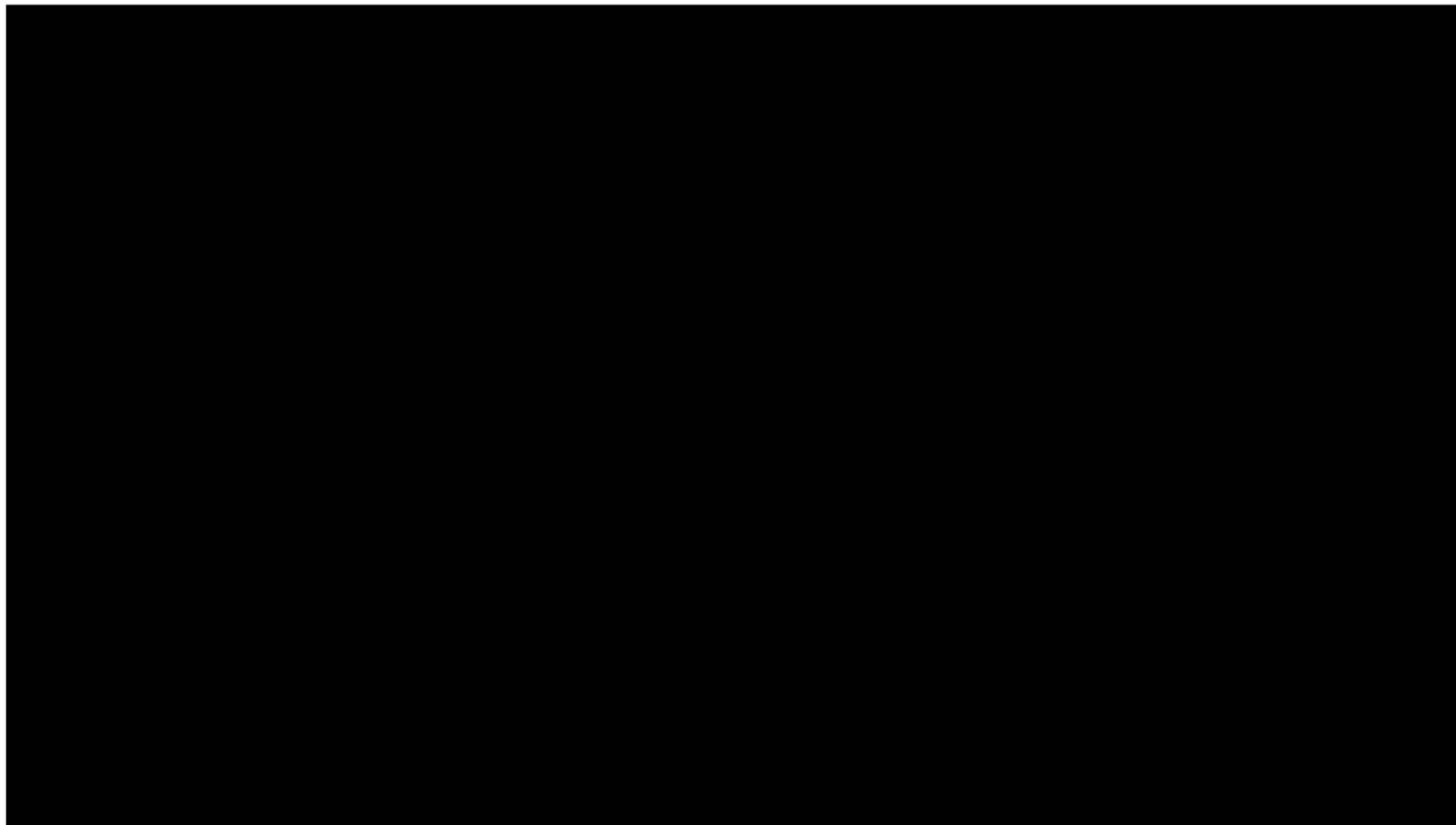
Other Applications

- Aerial Image Dehazing



Other Applications

- Underwater Image Dehazing



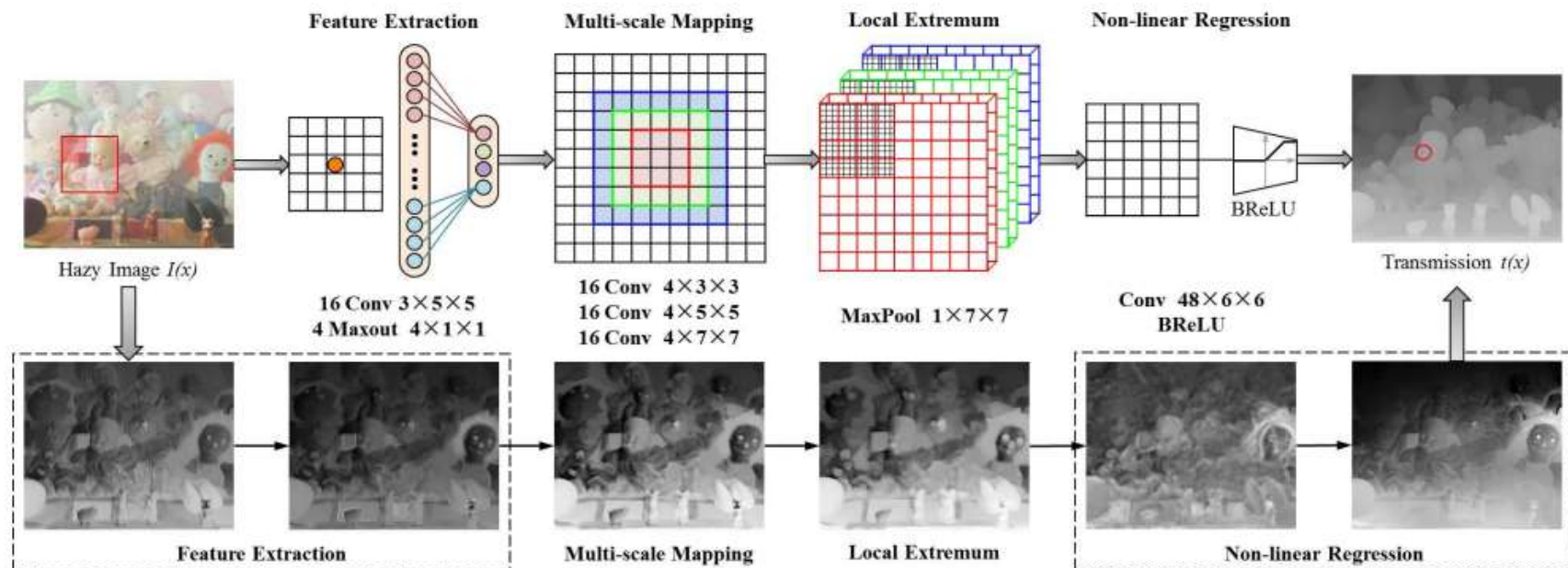
Learning-Based Approach

Learning-based Dehazing

- Methods based on the atmospheric scattering model
 - Early approach using CNN to estimate parameters of the ASM
- Methods using supervised learning
 - End-to-end learning independent of the ASM
- Methods using unsupervised learning
 - Not required synthetic data
- Methods using semi-supervised learning
 - Training with synthetic and real data

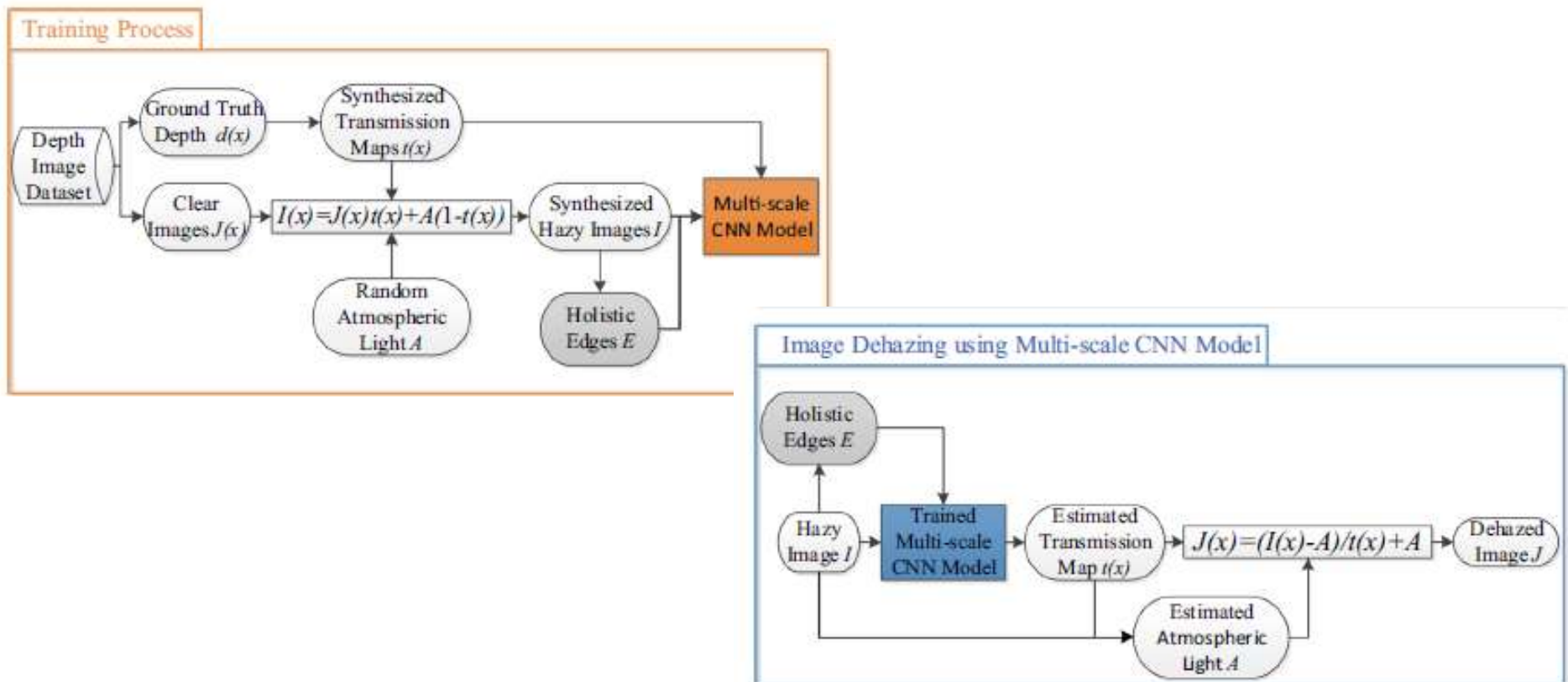
Methods Based on ASM: DehazeNet

- Takes a hazy image as input, and outputs its medium transmission map that is subsequently used to recover a haze-free image via atmospheric scattering model
- Synthesize training pairs of hazy and haze-free image patches/ select the highest intensity pixels as the atmospheric light



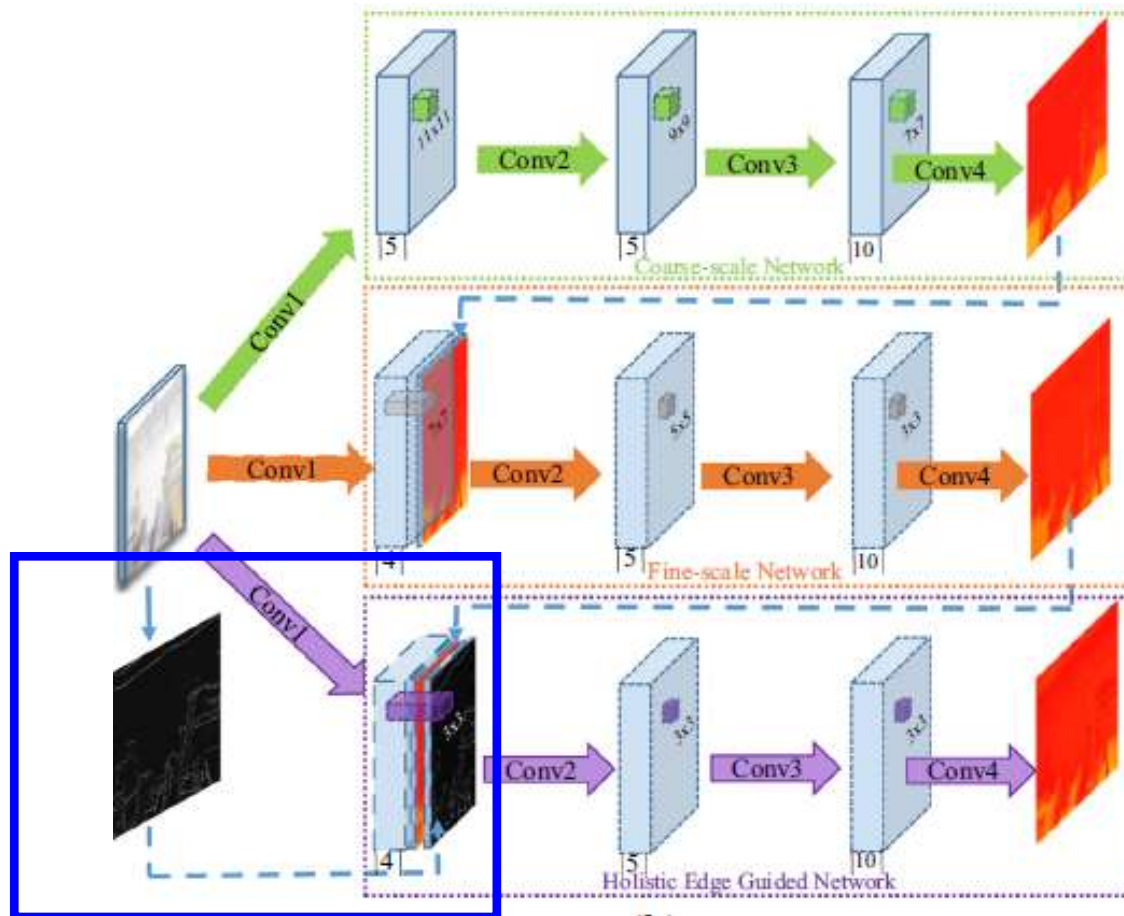
Methods Based on ASM: Holistic Edges

- Input: Haze images. Output: Transmission map
- Synthesize hazy images and the corresponding transmission maps based on a depth image dataset



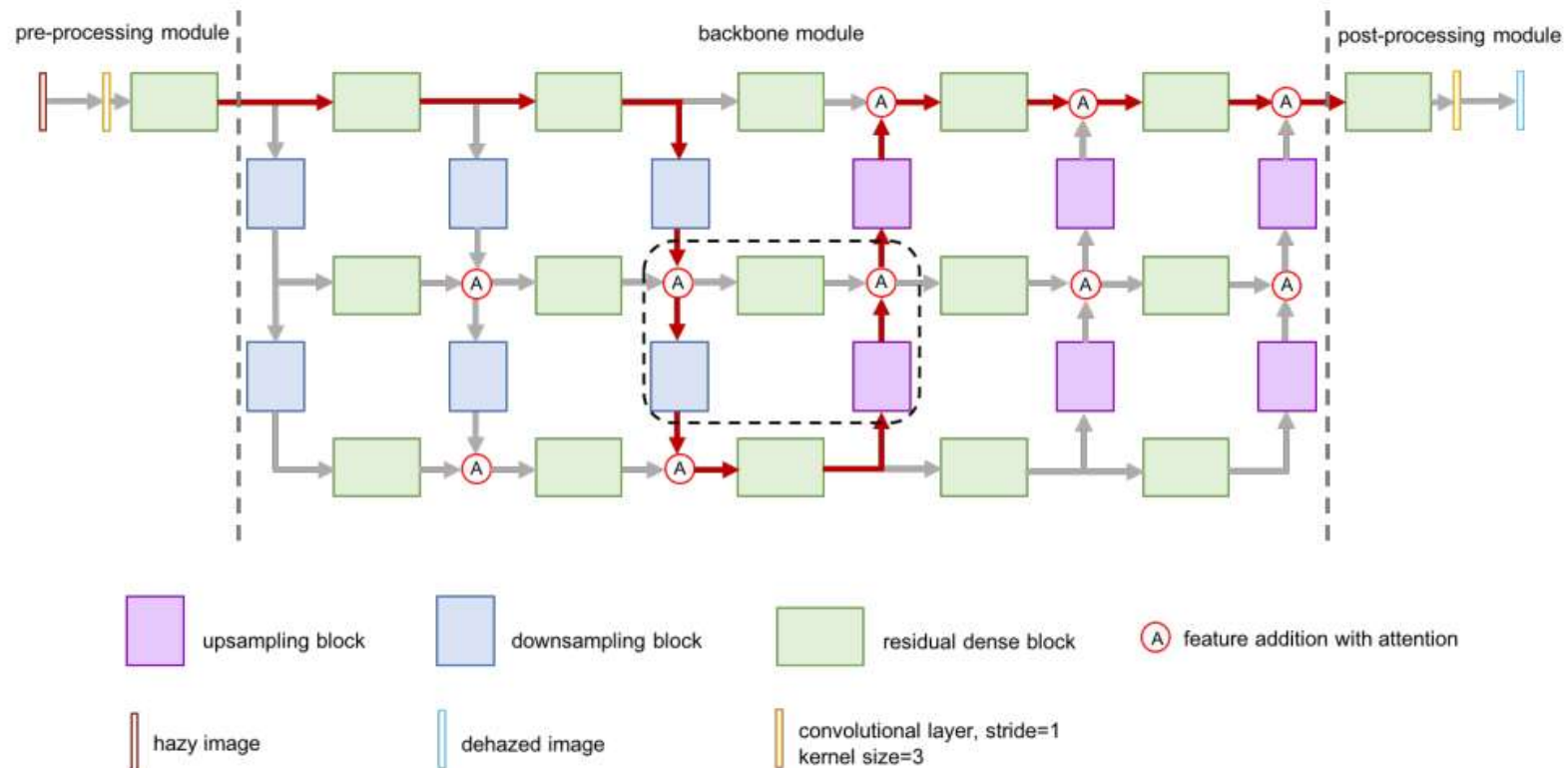
Methods Based on ASM: Holistic Edges

Using edge maps to refine the transmission map



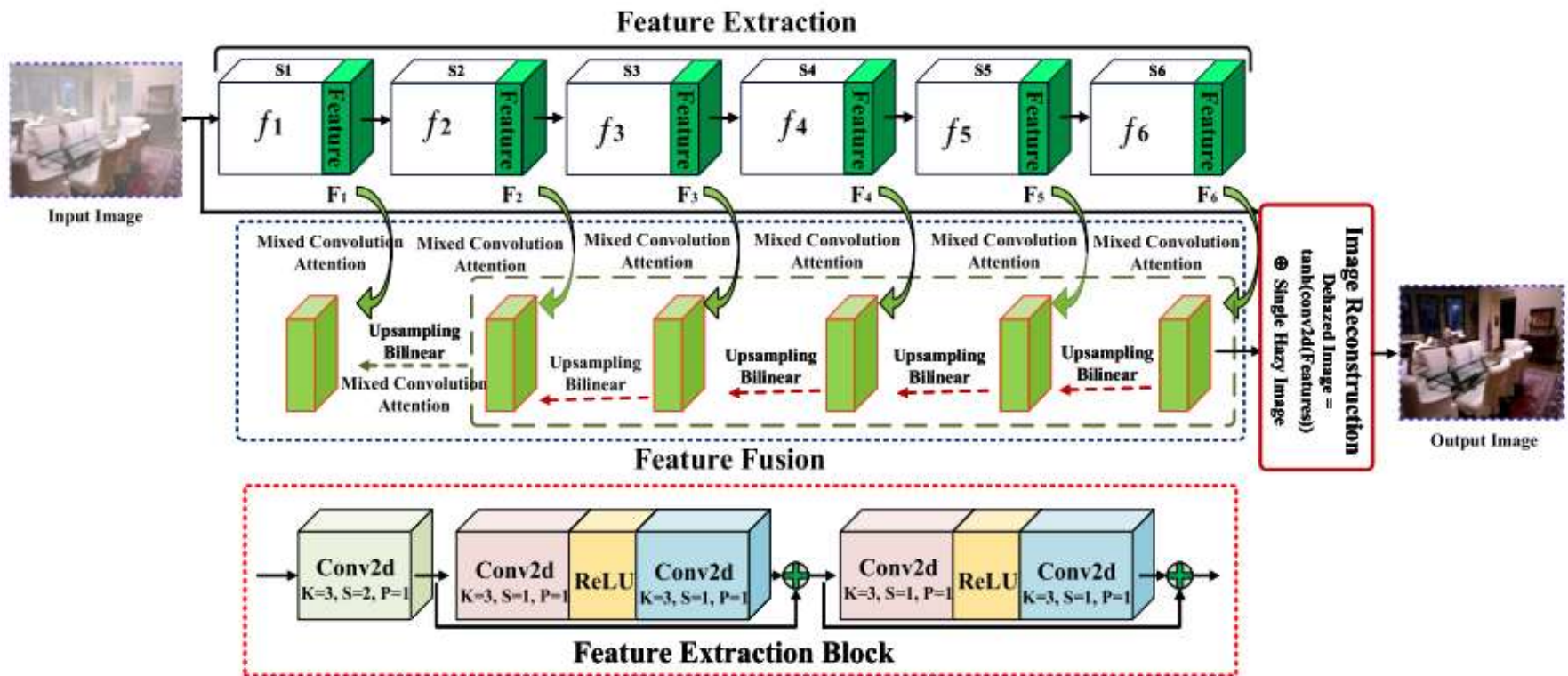
Using Supervised Learning: GridDehazeNet

- An end-to-end trainable CNN for a single image dehazing
- Model-agnostic approach that does not rely on the ASM
- Training and testing with synthetic datasets



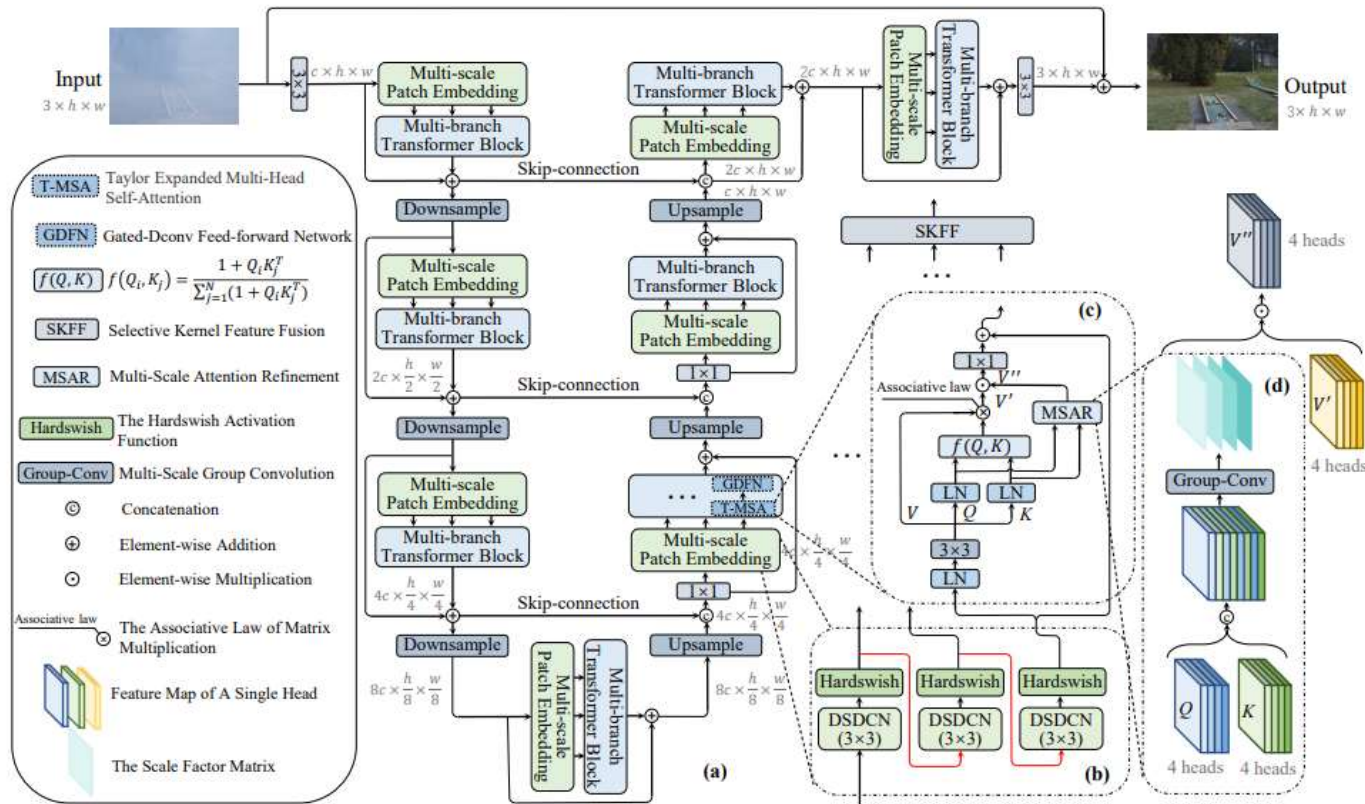
Using Supervised Learning: Hierarchical Feature Fusion

- An end-to-end network using multi-scale feature representations
- Training with synthetic dataset



Recent Approaches

- Attention-based/Transformer-based dehazing
- For example,



Using Semi-supervised/Unsupervised

- Supervised methods often require synthetic paired images that are inconsistent with real world hazy images.
- Using semi-supervised/unsupervised approaches

Using Unsupervised: Zero-Shot

- An unsupervised method that does not use any clean images as the ground-truth
- Use only the observed single hazy image to perform learning and inference



Using Semi-supervised

- Containing a supervised learning branch and an unsupervised learning branch
 - supervised branch: using synthetic images
 - unsupervised branch: using real hazy images
 - Sharing weights of two branches

